Transmission Volt VAR Control Connection Evidence and Commissioning

This companion to Volt VAR Droop Curve Design in Transmission Systems changes the main deliverable from a tuning method to an evidence package. The central question is whether the proposed and installed plant demonstrably meets its applicable voltage and reactive-control obligations. The engineering report explains curve behavior; this document shows how to turn that design into requirements, model evidence, tests and operating records.

The scope is transmission-connected resources, with Australian connection practice as the principal context and comparisons to European and North American frameworks. The process below is a proposed engineering assurance workflow. It does not declare a particular plant compliant or replace its registered performance standard, connection agreement or applicable rules.

1 Establish the applicable requirements

Begin with the resource type, jurisdiction, connection voltage, plant size, connection date, and whether the project is new, modified or existing. Obtain the applicable rule version, agreed access standard, connection agreement and operator instructions. Identify the physical point at which each obligation applies. A plant-terminal test may be inadequate for a requirement defined at the connection point.

Separate four categories in the requirements register: mandatory rules, obligations incorporated through an agreement, technical guidance, and project design preferences. Record the authority and version for each item. Do not convert a recommended capability range into an assumed operating setting, or label a design preference as a grid-code requirement.

For each requirement, state the physical quantity, base, measurement location, operating conditions and acceptance definition. A statement such as "4% droop required" is incomplete without its reactive normalization and convention. "Fast response" is incomplete without the event, onset definition, final target, settling band and treatment of saturation.

The register should identify unresolved applicability questions and assign them to the appropriate project role. Resolve them before finalizing the test plan. Otherwise, a technically successful test can produce evidence that does not answer the actual connection requirement.

2 Jurisdiction and standards comparison

Australian NER S5.2.5.13 explicitly accommodates agreed droop or reactive-current compensation and agreed regulation locations. A project must connect its curve definition to the applicable performance standard rather than interpret the regulation tolerance as a droop setting. The engineering report explains the relevant numerical distinction.[1]

European Regulation 2016/631 includes a defined slope convention and capability ranges for applicable power park modules. The compliance task is to identify the applicable module classification and the operator's selected settings and response criteria, including national implementation. A selectable range is not proof that any setting in that range meets the particular connection's performance needs.[2]

IEEE 2800-2022 addresses transmission IBR performance. IEEE 2800.2-2026, published on 30 June 2026, complements that framework with recommended conformity-assessment practices. The official catalogue describes equipment, conditions, tests, modelling methods and other assessment procedures. The full clauses and any adopted amendments are needed to build a clause-level compliance schedule; the public scope does not supply all numerical test criteria.[3]

Practical interpretation

Use a jurisdiction-specific register rather than a blended international checklist. A requirement drawn from another region can be a useful design benchmark, but should be labelled as such. When multiple documents apply, record how they interact and obtain an agreed interpretation for material differences in bases, locations or response definitions.

A compliance-focused report would therefore devote less space to deriving network sensitivities and more to traceability. It would identify exactly which result demonstrates which obligation and what evidence remains outstanding. This distinction also changes the language: "candidate gain" belongs to design development; "verified setting" requires completed evidence and an identified configuration.

The remainder of this document proposes records and tests that support that traceability. Their numerical acceptance values are deliberately project-specific, because no plant, agreement or approved settings have been supplied.

1. AEMC. National Electricity Rules S5.2.5.13 Voltage and reactive power control. Version-specific online text accessed September 2026.

2. European Commission. Commission Regulation EU 2016/631. 14 April 2016.

3. IEEE Standards Association. IEEE 2800.2-2026 Recommended Practice for Test and Verification Procedures for Inverter-Based Resources Interconnecting with Bulk Power Systems. Published 30 June 2026.

3 The design and model evidence package

AEMO's modelling requirements page identifies Power System Model Guidelines v3.0 and the 2025 design and setting data sheets. It describes the need for models that represent plant arrangement and performance across expected conditions. Use the current published framework and its associated submission documents to determine the required package for the connection.[4]

Evidence to prepare

The curve specification should contain voltage and reactive bases, sign conventions, reference and bias, directional slopes, deadbands, hysteresis, limits, and mode-dependent changes. Provide representative numerical points to eliminate ambiguity. Identify whether current or power is controlled and whether the reference moves with scheduled reactive output.

The capability evidence should show net connection-point limits over active power, voltage, equipment availability and relevant thermal or DC conditions. Include internal voltage and current constraints, shunt states and transformer taps. Record the source of each limit: OEM rating, calculation, test or agreed capability requirement.

The control description should map sensors to controlled quantities and show filtering, communication, PI action, anti-windup, limits, allocation and fallback logic. Match each model parameter to the installed controller parameter and record units. If a generic model cannot represent a material behavior, identify the limitation and the alternative evidence used to assess it.

Configuration consistency

Use one configuration identifier across the study report, model package, controller settings and commissioning record. Record software and firmware versions, model revision, checksum where practical, and approved deviations. A change to a filter or fallback mode can be material even if the headline droop percentage is unchanged.

Initialize models against the actual intended equilibrium. Confirm connection-point P, Q and V, internal states and active mode before applying disturbances. If the load-flow model uses ideal voltage control while the dynamic model uses permanent droop, reconcile the initial condition explicitly rather than accept an unexplained transient at simulation start.

4. AEMO. Modelling requirements. Living page; lists guidelines of 25 September 2025.

4 Studies and acceptance criteria

Begin with an agreed scenario matrix. Include dispatch and voltage schedules, relevant outages, weak and typical network conditions, resource availability and control-mode combinations. Each case should identify why it is included and which requirement it assesses. Avoid selecting cases solely because they are easy to solve or resemble the original tuning condition.

Static studies demonstrate feasible voltages, flows, capability and reserve under the assumed control states. Dynamic studies assess response and interactions. Detailed EMT studies are appropriate when the relevant mechanism depends on fast converter controls, waveform effects, unbalance or limiter transitions. The choice of model should be justified by the phenomenon being assessed.

Acceptance definitions to settle before testing

Specify the disturbance and its measurement point. Define whether rise time is measured from event onset or from observed response onset, and whether it refers to voltage, reactive current or reactive power. Define the final target under droop and the treatment of deadband and saturation. State the settling band and how long the response must remain inside it.

Set the applicable limits for overshoot, damping and sustained oscillation. Identify permitted transient excursions separately from continuous limits. Do not use a single generic tolerance for all variables or assume the curve's intended steady voltage offset is a regulation error.

For each simulation, retain the input case, model configuration, commands, waveforms, extracted metrics and automated checks. A pass/fail summary should link to supporting evidence and identify conditional results. If a candidate fails, classify whether the cause is capability, control dynamics, model inconsistency or the requirement interpretation before changing settings.

Review outcomes

An acceptable design package closes each applicable requirement with evidence or an explicit outstanding action. A solver convergence flag is not a performance result. An acceptable waveform under one network condition is not proof across the approved envelope. The final review should verify both individual criteria and consistency across all submitted models and tests.

5 Commissioning and mode transition tests

Commissioning should verify the implemented curve and the complete signal path. Confirm instrument scaling, current direction, voltage base, timestamp alignment and the active regulator before applying perturbations. Record connection-point quantities and sufficient internal signals to distinguish a commanded limit from a physical limit.

A static test sequence should cover both directions of response, deadband edges, reference changes and representative capability boundaries. Select steps large enough to exceed measurement uncertainty and small enough to remain within the approved test envelope. Fit the slope only to settled points from the same active segment and mode.

Dynamic tests should exercise the agreed small-disturbance criteria and relevant transitions. Include entry and exit from limiting, resource unavailability, control-mode transfer, primary-meter failure, backup-meter operation, stale measurements and communication restoration. Where site disturbance testing is unsuitable, agree the combination of simulation, controller integration testing and site verification that provides the required evidence.

Post-test restoration

AEMO's report on the June 2022 South Australian oscillations connects a control-code error to a communication fail-safe test and incorrect subsequent control behavior. It makes restoration checks a concrete engineering concern, not an administrative formality. The incident is described in more detail in the first report.[5]

After each test, verify the active control mode, final references, limits, code and settings. Confirm that alarms clear for the correct reason and that normal voltage response has returned. Preserve the test configuration and restored configuration separately, with a recorded comparison.

Continue monitoring after the test sequence ends. A delayed instability or an incorrectly restored supervisory mode may not appear during a short inspection. Define who reviews the post-test record, the trigger for stopping further tests and the process for returning the plant to an agreed condition.

The commissioning conclusion should identify the tested configuration and scope. Avoid a general statement of compliance if the evidence covers only one operating mode or a limited capability range.

5. AEMO. Power System Oscillations in South Australia on 23 June 2022. February 2023.

6 Handover change control and reusable records

The operating handover should explain which modes are available, which settings operators may change, what reserve or capability alarms mean, and what happens after loss of communication. Include the path for reporting oscillatory behavior and obtaining a controlled settings review. Keep the approved voltage schedule and curve convention visible in the operating instructions.

A change-impact assessment should ask whether modified firmware, added resources, revised compensation, new network topology or nearby control changes invalidate previous evidence. Repeat the studies and tests that address the changed mechanism. A documented assessment is more useful than either assuming every change is harmless or rerunning every historical test indiscriminately.

Requirement record

For each item, record a unique identifier, source and version, applicability rationale, required capability or performance, measurement location and units, operating envelope, acceptance criterion, evidence method, evidence file, tested configuration, reviewer and status. Distinguish "not applicable," "not assessed," "conditional" and "passed." These statuses represent different conclusions.

Test record

Record the objective and requirement identifiers, prerequisite condition, approved perturbation, expected response, abort criteria, measured channels, sampling and time alignment, controller and model versions, results, deviations and restoration checks. Include enough information for another engineer to repeat the analysis from the retained data.

Example of a complete claim

An evidence statement can say that a specified curve segment was measured at the agreed point, for a named active-power range and equipment configuration, and met its documented slope tolerance using a stated fitting method. That is stronger than saying simply that "droop works." Equally, it should not claim fault-response capability when only normal-operation steps were tested.

The final output of a compliance emphasis is a traceable connection and operating record. It follows the engineering report naturally: curve design creates the candidate behavior; the evidence package establishes what was demonstrated, under which conditions, and for which installed configuration.

Sources

Sources accessed 13 September 2026. The proposed records and workflow are engineering recommendations. The source provisions do not determine applicability to an unidentified connection, and full normative IEEE test clauses are not reproduced from catalogue summaries.

1. AEMC. National Electricity Rules S5.2.5.13 Voltage and reactive power control. Version-specific online text accessed September 2026. Connection-specific applicability must be established.

2. European Commission. Commission Regulation EU 2016/631. 14 April 2016. Articles 2 and 21; classification and national implementation matter.

3. IEEE Standards Association. IEEE 2800.2-2026 Recommended Practice for Test and Verification Procedures for Inverter-Based Resources Interconnecting with Bulk Power Systems. Published 30 June 2026. Official scope and status; complements IEEE 2800. Full normative procedures not reproduced.

4. AEMO. Modelling requirements. Living page; lists guidelines of 25 September 2025. Model Guidelines v3.0 and 2025 design and setting data sheets.

5. AEMO. Power System Oscillations in South Australia on 23 June 2022. February 2023. Overview and Appendix A2; communication fail-safe and restoration behavior.